$[OH^-] = 1 \times 10^{-4}$ mol litre$^{-1}$

$pOH = -\log_{10}[OH^-] = -\log_{10} \times [10^{-4}]$

$= -(-4 \times \log_{10}10) = -(-4) = 4$

Since, $pH + pOH = 14$

$pH = 14 - pOH = 14 - 4$

$= 10$

**Example 4:** Calculate the pH of a solution in which the concentration of the hydrogen ions is $1.0 \times 10^{-8}$ mol litre$^{-1}$.

**Solution:** Here, although the solution is extremely dilute, the concentration given is not of an acid or a base but that of $H^+$ ions. Hence, the pH can be calculated from the relation:

$pH = -\log_{10}[H^+]$

given $[H^+] = 1.0 \times 10^{-8}$ mol litre$^{-1}$

$pH = -\log_{10}10^{-8} = -(-8 \times \log_{10}10)$

$= -(-8 \times 1) = 8$

**Example 5:** If the pH of a solution is 4.5, what is its pOH?

**Solution:**

$pH + pOH = 14$

$pOH = 14 - 4.5 = 9.5$

$pOH = 9.5$

## Points to Remember

- A chemical change is a change in which one or more new substances are formed.
- Most combination reactions are exothermic
- All photo decomposition reaction are endothermic reactions.
- Double displacement reaction or metathesis may occur by the mutal exchange of ions.
- Precipitation reaction gives an insoluble salt as the product.
- Neutralisation reactions are reactions between an acid and a base that forms salt and water.
- Neutralisation prevents tooth decay.
- Most reactions in chemistry are irreversible reactions.
- Chemical equilibrium-the rate of the forward reaction is equal to rate of the back ward reactions.
- Equilibrium is possible in a closed system.
- Temperature increases the reaction rate.
- Pressure increases the reaction rate.
- The term pH means power of hydrogen.
- pH plays a vital role in everyday life.
- In humans all bio chemical reactions take place between the pH value of 7.0 to 7.8.
- If pH of rain water is below 5.6 its called acid rain.
- Pure water is a weak electrolyte.



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